EXP NO:12 Experiment to detect outliers in a given data set.

AIM:
To understand the procedure to identify the outliers in a given data set.

PROGRAM:

python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Creating dataset of student marks
data = {'Marks': [45, 52, 48, 50, 55, 47, 51, 49, 98, 3, 53, 46]}
df = pd.DataFrame(data)
df

# Statistical description
df.describe()

# Boxplot to visualize outliers
sns.boxplot(x=df['Marks'])
plt.title("Boxplot of Marks")
plt.show()

# Calculate quartiles and IQR
Q1 = df['Marks'].quantile(0.25)
Q3 = df['Marks'].quantile(0.75)
IQR = Q3 - Q1

# Calculate lower and upper limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

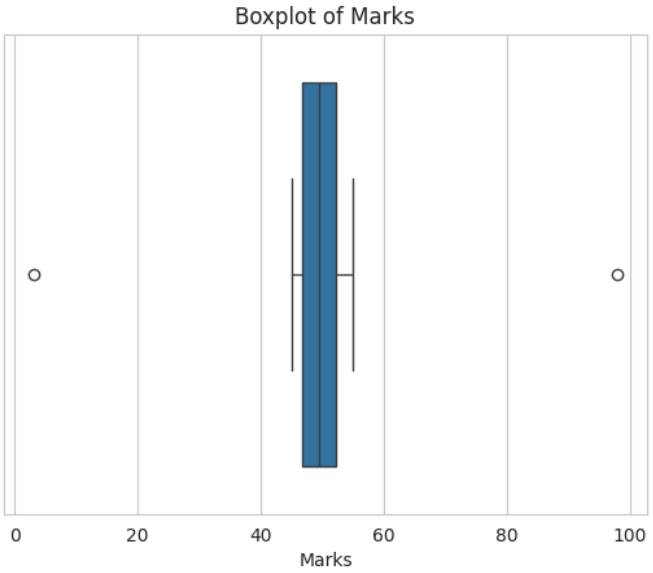
print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

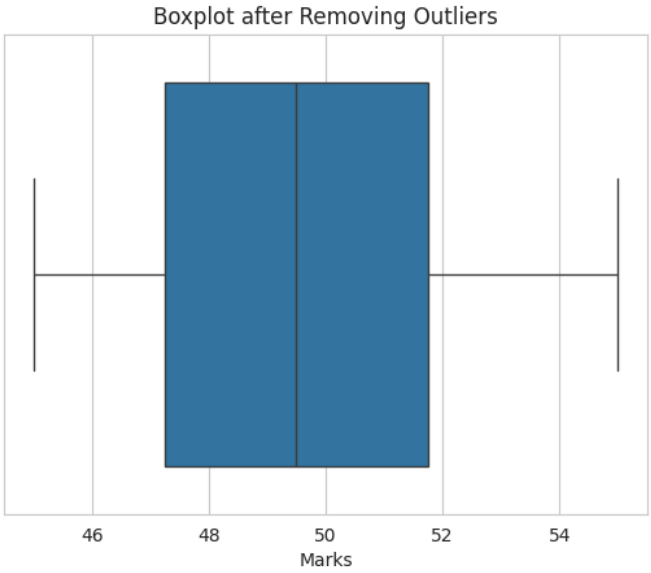
# Detect outliers
outliers = df[(df['Marks'] < lower_limit) | (df['Marks'] > upper_limit)]
print("Outliers:")
print(outliers)

# Remove outliers
cleaned_df = df[(df['Marks'] >= lower_limit) & (df['Marks'] <= upper_limit)]
cleaned_df

# Boxplot after removing outliers
sns.boxplot(x=cleaned_df['Marks'])
plt.title("Boxplot after Removing Outliers")
plt.show()
```

OUTPUT:





RESULT:

Thus the program to identify the outliers in the given data set using the IQR method was executed successfully.